8/03

Doubts on DMA

int a[10];// static memory

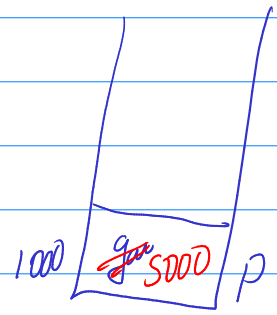
memory allocated is fixed
memory shortage
memory wastage

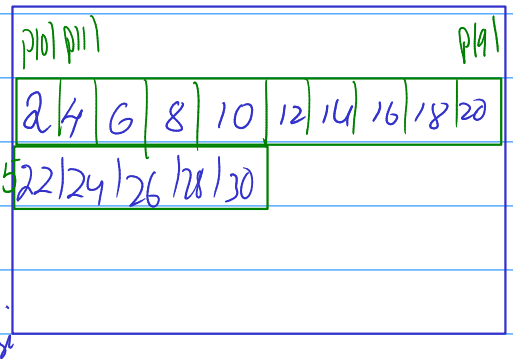
To over come this we go for Dynamic memory allocation.
In C we have predefined functions for DMA.

1. malloc
2. calloc
3. realloc
4. free

} DMA
→ resizing
→ deallocation

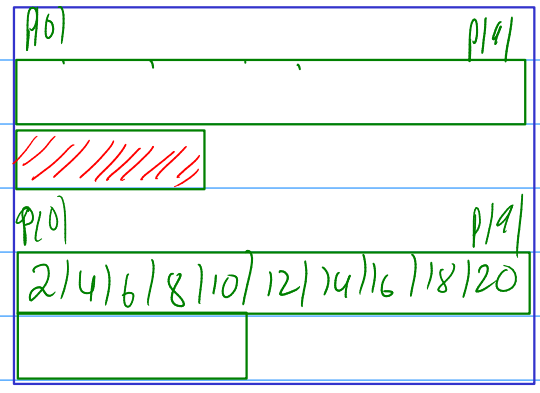
```
int main()
{
int size=10;
int *p;
p=malloc(size*sizeof(int));
for(i=0; i<size; i++)
{ printf("enter the elem");
  scanf("%d",&p[i]);
}
```






```
for(i=0; i<size; i++)
{ printf("%d", p[i]);
```

p = malloc((size+5)*sizeof(int));
malloc, calloc cannot resize
we need to use realloc




```
p = realloc(p, (size+5)*sizeof(int));
for(i=10; i<15; i++)
{
```

P=NULL

Add(int *P)

{ P=realloc(P,(cnt+1)*sizeof(int));

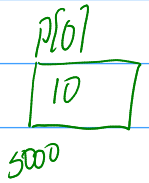
scf("%d",&P[cnt]);

Add(); cnt++;

cnt=0

if P is NULL, then its a choice of realloc where user it want it can return

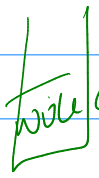




int main()

{ int *a=NULL;

can 'A': Add(a);

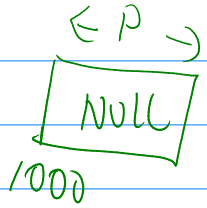


Main()

{ int *P=NULL;

f(P);

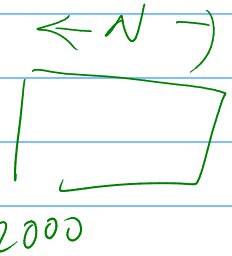
}



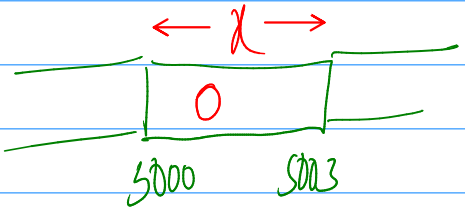
int f(int *N)

{

}



int x=0;



name of blk x

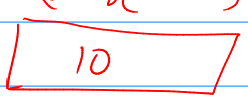
Add of x 5000

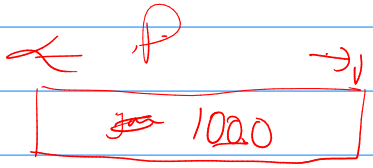
Value at 0

int *P;

int x=10

P=&x;





P → 1000

&P → 2000

*P → Value at P → Value at 1000 → 10